

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 25)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

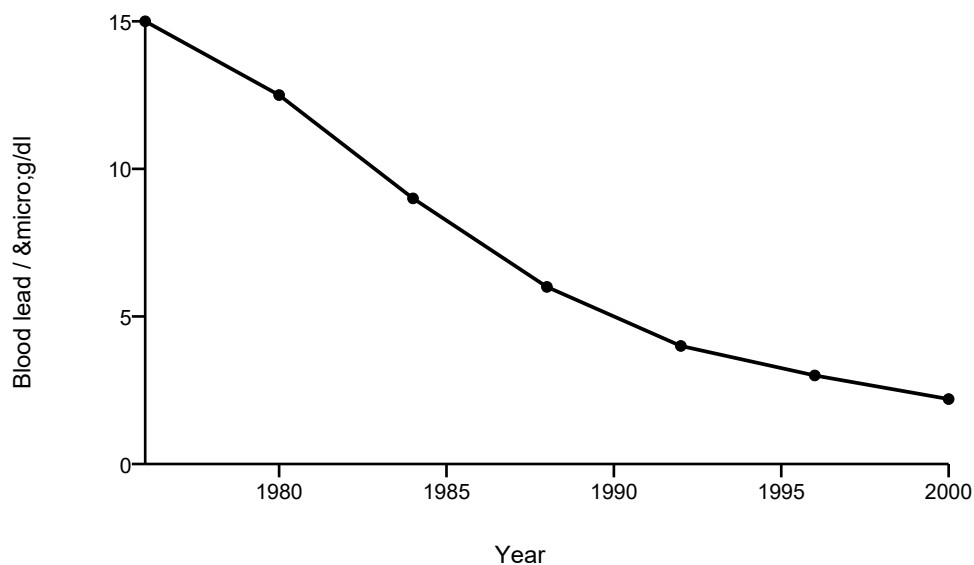
Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1

Figure 1 shows the mean concentration of lead in the blood of children in a large city between 1976 and 2000, and the year in which lead additives in petrol began to be phased out.

Figure 1



0 1

1 Calculate the percentage decrease in mean blood lead between 1976 and 2000.

[2 marks]

Answer

%

0 1

2 Explain why tetraethyl lead was added to petrol and why this caused a public health threat.

[3 marks]

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0 1 . 3

Suggest **two** reasons, other than petrol additives, why blood lead concentrations continued to fall after 1990.

[2 marks]

1

2

0 1 . 4

Explain how adding small amounts of phosphoric acid to a water supply reduces lead in drinking water.

[2 marks]

0 1 . 5

Explain why discarded lead fishing weights and shotgun pellets can poison waterfowl.

[1 mark]

10

0 2

A pesticide was applied to a field of oilseed rape. Table 1 shows the concentration of the pesticide in organisms sampled from a pond next to the field.

Table 1

Organism	Trophic level	Pesticide concentration / mg kg^{-1}
Pond water	-	0.002
Phytoplankton	1	0.04
Water fleas (Daphnia)	2	0.5
Small fish	3	4.2
Heron	4	38.0

0 2 . 1

Calculate how many times more concentrated the pesticide is in the heron than in the pond water. Give your answer to two significant figures.

[2 marks]**Answer**

times

0 2 . 2

Name the process shown by the data in Table 1.

[1 mark]

0 2 . 3

Explain the **two** properties of the pesticide that cause the pattern shown in Table 1.

[4 marks]

0 2 . 4

The pesticide was probably an organochlorine. Suggest why the use of organochlorines is now banned or heavily restricted in most countries, even though DDT is still used to control malaria mosquitoes in some houses.

[3 marks]

10

0 3 A student investigated the effect of pH on the germination of cress seeds.

0 3 . 1 Explain why cress is a suitable species for this investigation but the results may not apply directly to natural ecosystems.

[2 marks]

0 3 . 2 Describe how the student should decide the range of pH values to test.

[2 marks]

0 3 . 3 Give **three** variables that should be standardised and explain how long the experiment should continue.

[4 marks]

0	3	.	4
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The student wants to compare the effect of nutrient concentration on the growth of duckweed. Explain why leaf area is a better measure of growth than fresh mass.

[2 marks]

10

0	4	.	1
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Evaluate the extent to which new technologies could allow fossil fuels and nuclear fission to continue to supply a large proportion of the world's energy sustainably.

[9 marks]

Extra space

04 . 2

State the approximate percentage of natural gas in a reservoir that can be recovered using its own pressure.

[1 mark]

10

0	5	.	1
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Discuss how the features of energy resources determine their suitability for different uses and their environmental sustainability.

[9 marks]

Extra space

05 . 2

State **one** reason why nuclear power stations can be located where transport infrastructure is poor.

[1 mark]

10

06 . 1

Which of the following best describes the purpose of Critical Group Monitoring? Tick one box.

[1 mark]

- A To detect health damage in the population most affected by a pollutant ☒
- B To predict the route a pollutant will take through the environment ☐
- C To assess exposure of the people most at risk so that everyone else ☒ can be assumed to be safe ☐
- D To measure the concentration of a pollutant at the point of release ☐

06 . 2

Explain why CPA can reduce the cost of an environmental monitoring programme.

[2 marks]

06 . 3

A new chemical has properties similar to PCBs (persistent, liposoluble chlorinated organic compounds). Explain how knowledge of PCBs helps decisions about the new chemical.

[2 marks]

5

07

Table 1 shows information about two copper deposits, X and Y, being assessed for mining. The current cut-off ore grade for copper is 0.4%.

Table 1

Feature	Deposit X	Deposit Y
Ore grade / % copper	0.9	0.35
Depth of deposit / m	350	40
Overburden	Hard igneous rock	Loose glacial sediment
Distance to nearest railway / km	120	8
Annual rainfall / mm	2 100	500
Chemical form	Copper sulfide	Copper sulfide

0	7	.	1
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Explain what is meant by the **cut-off ore grade** and state which deposit is currently included in the copper reserves.

[2 marks]

0	7	.	2
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Use Table 1 to discuss the factors that affect the viability of mining each deposit.

[6 marks]

0	7	.	3
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State **two** changes that would decrease the cut-off ore grade.

[2 marks]

10

08 . 1

Which of the following statements about stratospheric ozone is correct? Tick one box.

[1 mark]

- A Ozone is destroyed only by CFCs ☐
- B Ozone absorbs mainly infrared radiation ☐
- C Ozone is formed when monatomic oxygen reacts with diatomic oxygen ☐
- D Ozone concentrations are highest at ground level ☐

08 . 2

Explain the difference between the roles of ozone in the stratosphere and in the troposphere.

[2 marks]

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08 . 3

Explain why HCFCs cause less ozone depletion than CFCs.

[2 marks]

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5

09 . 1

Explain why phosphorus is more likely than nitrogen to be a limiting factor on biological productivity.

[3 marks]

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09 . 2

Explain how mycorrhizal fungi increase phosphate uptake by plants.

[2 marks]

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09 . 3

Describe how human activities have altered the phosphorus cycle.

[3 marks]

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09 . 4

Explain **two** strategies for the sustainable management of phosphorus supplies.

[2 marks]

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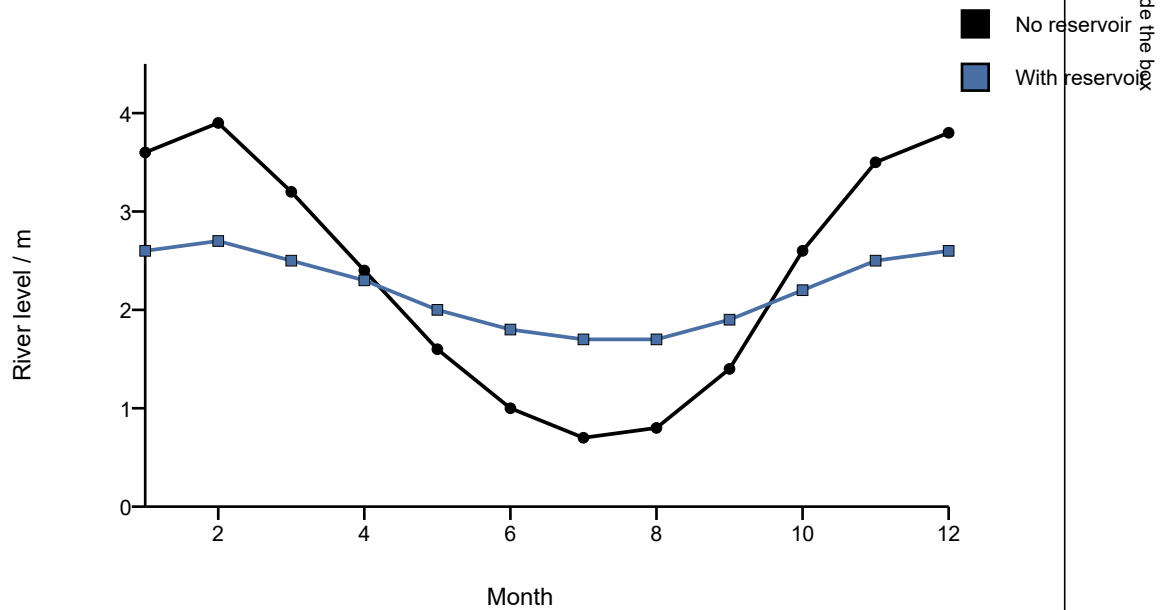
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1 0

Figure 1 shows the river level below a reservoir site during one year, before and after a river-regulation reservoir was built.

Figure 1



1 0

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Use Figure 1 to calculate the reduction in the annual range of river level produced by the reservoir.

[2 marks]

Answer

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1 0

2

Explain how the reservoir produces the changes shown in Figure 1.

[2 marks]

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1 0

3

Explain why human activities in the catchment may have increased the extremes in river flow before the reservoir was built.

[2 marks]

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10 . 4

Discuss the factors that should be considered when selecting a site for a new reservoir, and evaluate the alternative strategies for increasing the sustainability of water supplies.

[9 marks]

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Extra space

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15

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations.

[25 marks]

OR

1 1 . 2

Discuss how an understanding of the hydrological cycle can be used to develop strategies for the sustainable management of water resources.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

[illegible]

[illegible]

END OF QUESTIONS